

COURSE 3 - DATA ENGINEERING 2026**C3-05 Targeted Remediation Cards**

Repair only the failed distributed-processing competency, then prove it on a changed micro-case.

Artifact	C3-05-REPAIR
Status	2026 - LEARNER EDITION
Phase	C3-05 Distributed Processing with Spark 16 lessons 46-60 focused hours
Prerequisite	C3-04 Gate PASS; C2B SQL/Python/modeling foundations assumed, distributed execution taught here.
Current lane	Apache Spark / PySpark 4.2.0 Python 3.10+ Java 17+ learner local[2]
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny worked example -> guided real Spark workflow -> expected result -> why it worked -> break/fix -> independent changed task -> validation -> physical-plan/production reasoning -> evidence. A green Spark job is not mastery if grain, quality, plan or reproducibility is wrong.

Repair rules

Do not restart the whole phase for one narrow weakness. Preserve the failed evidence, use the matching card, close review and solve a changed micro-case. Completing repair never assigns Gate B/C automatically.

R-ARCH - Execution model

Redraw a changed pipeline and correctly label driver, partitions, tasks, stages, action and shuffle.

PASS: changed evidence is correct, reproducible, explainable and completed with the review closed.

R-SCHEMA - Schema + quality

Use a fresh CSV with one malformed numeric field; explicit schema/raw capture + clean/reject reconciliation required.

PASS: changed evidence is correct, reproducible, explainable and completed with the review closed.

R-JOIN - Join grain

Use a new fact/dimension case with one duplicate dimension key; block the join before fan-out, then repair and prove grain.

PASS: changed evidence is correct, reproducible, explainable and completed with the review closed.

R-GRAIN - Aggregation/window grain

Build one groupBy and one window metric on a changed domain with explicit grains and deterministic tie rule.

PASS: changed evidence is correct, reproducible, explainable and completed with the review closed.

R-PART - Partition/shuffle

Use a new hot-key distribution and compare repartition/coalesce from evidence; no coalesce(1) shortcut.

PASS: changed evidence is correct, reproducible, explainable and completed with the review closed.

R-PLAN - Plan diagnosis

Annotate a new formatted plan and connect each Exchange/join/aggregate node to code/data cause.

PASS: changed evidence is correct, reproducible, explainable and completed with the review closed.

R-SKEW - Skew

Measure a new key distribution and choose the least-invasive remedy after classifying valid vs dirty concentration.

PASS: changed evidence is correct, reproducible, explainable and completed with the review closed.

R-TEST - Testing

Write fresh DataFrame/schema/invariant tests for a changed transformation.

PASS: changed evidence is correct, reproducible, explainable and completed with the review closed.

R-HANDOFF - Repeatable job

Parameterize a new input/output path, capture version/command/validation and hand off via README/runbook.

PASS: changed evidence is correct, reproducible, explainable and completed with the review closed.